higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N6

20 NOVEMBER 2019

This marking guideline consists of 8 pages.

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QUESTION 1: COMPUTER SYSTEMS

1.1
1.1.1 C
1.1.2 B
1.1.3 C
1.1.4 C

(4 × 1) (4)

1.2
1.2.1 True
1.2.2 True
1.2.3 False
1.2.4 True

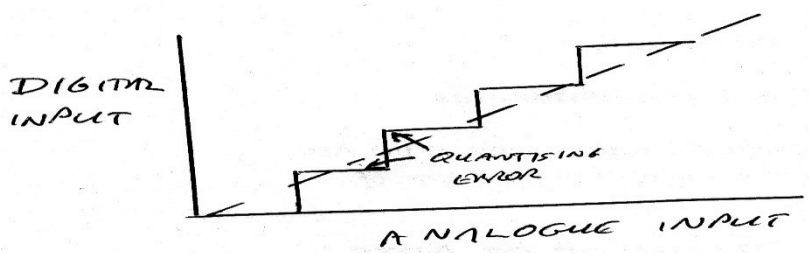
(4 × 1) (4)

1.3
1.3.1 E
1.3.2 F
1.3.3 D
1.3.4 B

(4 × 1) (4)

1.4
- Cost of a new system or expansion of the existing system
- Hiring of additional and specialised personnel
- Training of personnel
- Advantages and benefits that can be derived from the proposed system
- Environmental considerations
- Problem areas as well as possible solutions
- Commissioning and installation
- Service and backup facilities
- Data files and format requirements at both input and output terminals
- Future expansion and estimated lifetime of the system (Any 6 × 1) (6)

1.5



(2)
[20]

QUESTION 2: TRANSMISSION, DATA ACQUISITION AND RELATED HARDWARE

2.1
2.1.1 D
2.1.2 C
2.1.3 C
2.1.4 C

(4 × 1) (4)

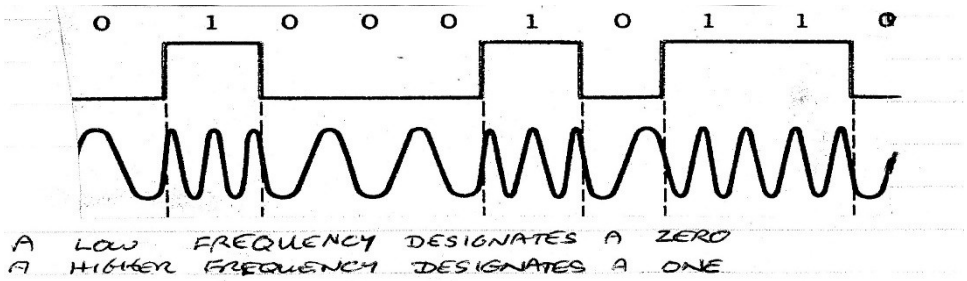
2.2
2.2.1 False
2.2.2 True
2.2.3 True
2.2.4 True

(4 × 1) (4)

2.3
2.3.1 B
2.3.2 A
2.3.3 D
2.3.4 C

(4 × 1) (4)

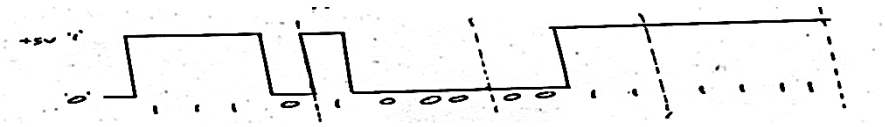
2.4 **NOTE**: Any wave train can be drawn so long as the frequencies on the 1 are visibly higher than the frequencies on the 0 and these frequencies must be a constant amplitude throughout.



(3)

2.5 Pulse distribution (1)

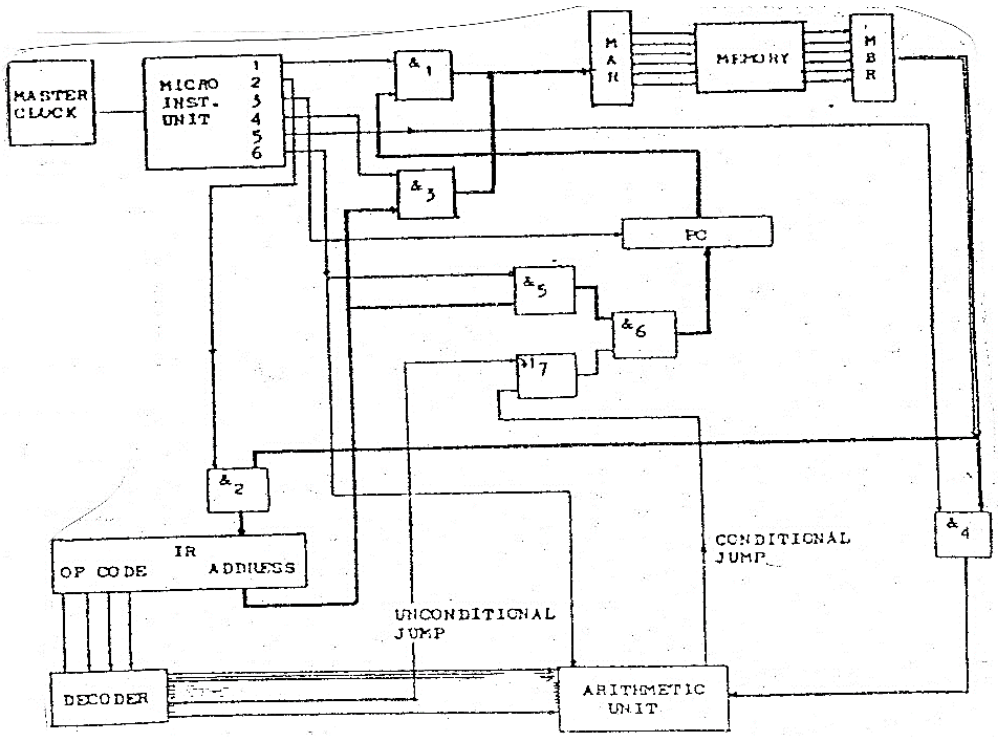
2.6



(4)
[20]

QUESTION 3: COMPUTER ARCHITECTURE

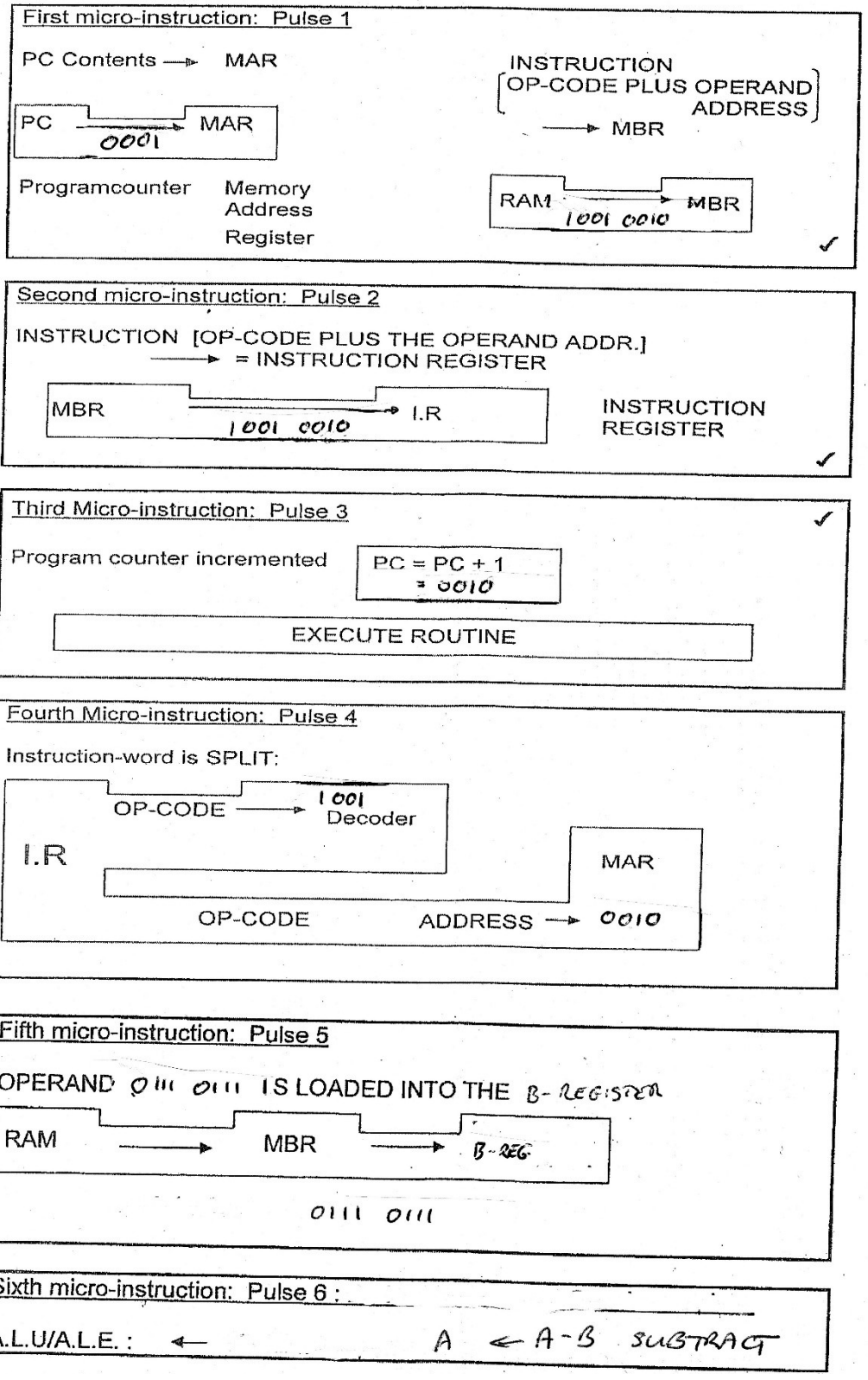
3.1



(10)

3.2

FETCH ROUTINE



(10)
[20]

QUESTION 4: HIGH-LEVEL PROGRAMMING

4.1

PASS	PINK	FLOYD	RESULT
0	8	8	16
		15	
- - - - -	- - - - -	- - - - -	- - - - -
1			23
		22	
- - - - -	- - - - -	- - - - -	- - - - -
2			30
		29	
- - - - -	- - - - -	- - - - -	- - - - -
3			37

8
29
37 (10)

NOTE:

1. The column PASS can start on 1 and not 0.

2. Each correct row (to the dashed line – which does NOT have to be included) is worth TWO marks – NO half marks. Mistakes must NOT be followed through.

3. The final printout below the table must be in the correct order, one below the other for TWO marks.

4.2 Stack pointer
19

Stack:

Address	Contents
16	04_{16}
17	23_{16}
18	$F1_{16}$
19	$2B_{16}$

NOTE: ONE mark for incrementing the stack pointer, ONE for indicating the new address and ONE for including the new contents. If the candidate did NOT redraw the rest of the stack, only the mark for the correct stack pointer can be given. (3)

4.3
- Salaries and wages
- Bills
- Examination results (Any relevant answer) (1)

4.4

INSTRUCTION NUMBER	ACCUMULATOR CONTENTS
0	0011 0011
01	0111 0111
02	1111 1111
03	1000 1000
04	0111 0111

NOTE: The instruction number may also be given in binary. (5)

4.5 Machine code (1)
[20]

QUESTION 5: NUMBER SYSTEMS

5.1 $1_1\ 0_2\ 1_3\ 0_4\ 1_5\ 1_6\ 1_7\ 0_8\ 1_9\ 1_{10}$

Pos. 1 checks 3; 5; 7; 9
1 1 1 1 – P1 should thus be 0: NOT thus: 1

Pos. 2 checks 3; 6; 7; 10
1 1 1 1 – P2 should thus be 0: IT IS thus: 0

Pos. 4 checks 5; 6; 7
1 1 1 – P4 should thus be 1: NOT thus: 1

Pos. 8 checks 9; 10
1 1 – P8 should thus be 0: IT IS thus: 0

Thus the fault lies on bit $0101_2 – 5_{10.}$

Thus pos. 5 which is a 1 should be a 0.

The word should be: **$1010\underline{0}11011_{hamming}$** (10)

5.2 $+0{,}10100000 \times 10^{+000}$

$= 0{,}101_2$

$= 0{,}5 + 0{,}125$

$= 0{,}625_{10}$ (3)

5.3 **$0101\ 1000\ 0100_{8.4.2.1}$**

NOTE: ONE mark for each nibble (4-bit string). If the subscript is omitted, the answer is wrong. (3)

5.4 **1000101_{gray}**

NOTE: ONE mark for the correct conversion, ONE for showing the subscript 'gray'. (2)

5.5

$$\overline{A + B + C + D} = \bar{A} \cdot \bar{B} \cdot \bar{C} \cdot \bar{D}$$

$$\overline{A \cdot B \cdot C \cdot D} = \bar{A} + \bar{B} + \bar{C} + \bar{D}$$

(2)
[20]

TOTAL: 100

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